

INFOSYS SPRINGBOARD INTERNSHIP PROJECT REPORT

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BATCH NO : 1

PROJECT CODE : #M-1-9

**PROJECT TITLE : DEVELOPMENT OF AI RESPONSE VALIDATION
SYSTEM WITH HALLUCINATION DETECTION
ASSISTANCE - GROUP 2**

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Development of AI Response Validation System with Hallucination Detection Assistance – Group 2

1. INTRODUCTION

1.1 Background

Large Language Models (LLMs) are increasingly used for generating natural language responses in applications such as question answering, education, content generation, customer support, and information retrieval. Although these models can produce fluent and informative responses, they may sometimes generate incorrect, incomplete, irrelevant, or unsupported information. Such incorrect information is commonly referred to as hallucination.

Traditional evaluation of LLM responses often depends on manual comparison with reference answers. This process can be time-consuming and may not provide a consistent evaluation across different responses. Therefore, an automated system is required to assess the quality of AI-generated responses using multiple evaluation criteria.

The proposed Development of AI Response Validation System with Hallucination Detection Assistance addresses this requirement by combining Retrieval-Augmented Generation (RAG), vector-based retrieval, ChromaDB, embedding generation, and multiple validation agents. The system retrieves relevant reference information and evaluates an AI response based on accuracy, relevance, hallucination, and completeness.

1.2 Project Overview

The developed system provides an automated environment for validating AI-generated responses against reliable reference information. The user can provide a question, an AI-generated response, and an optional reference document. The reference information is processed and converted into numerical embeddings using the all-MiniLM-L6-v2 embedding model.

The generated embeddings are stored and searched using ChromaDB, which enables the system to retrieve the most relevant reference information for a given question. The retrieved reference and AI-generated response are then passed to different validation agents.

The system produces individual scores for accuracy, relevance, hallucination, and completeness. These scores are combined to generate an overall score and final verdict. The

results are displayed through an interactive dashboard and can also be generated as a report for further analysis and documentation.

1.3 Need for the System

The system is developed to address the following challenges:

- Manual evaluation of AI responses requires considerable time.
- AI-generated responses may contain hallucinated or unsupported information.
- A response can be relevant but still factually incorrect.
- Important information may be missing from an otherwise correct response.
- Different evaluation criteria need to be considered independently.
- Evaluation results should be recorded and presented in an organized manner.
- Large numbers of responses require an automated batch evaluation mechanism.

The proposed system provides a structured approach to address these challenges through automated retrieval and multi-criteria validation.

1.4 Problem Statement

Large Language Models can generate responses that appear fluent and informative but may contain factual errors, irrelevant information, unsupported claims, or missing important details. Identifying these issues manually is difficult, especially when a large number of AI-generated responses need to be evaluated.

Existing evaluation approaches may focus on only one aspect of response quality, such as correctness or similarity, and may not provide a complete analysis of the response. There is therefore a need for an automated validation system that can compare an AI-generated response with reliable reference information and evaluate it using multiple quality criteria.

The proposed system addresses this problem by retrieving relevant reference information using a Retrieval-Augmented Generation (RAG) workflow and evaluating the response through separate validation agents for accuracy, relevance, hallucination, and completeness. The system generates individual scores, explanations, an overall score, and a final verdict to provide a structured assessment of AI response quality.

1.5 Scope of the Project

The scope of the project includes:

- Accepting user questions and AI-generated responses through an interactive interface.
- Supporting reference information through PDF documents and datasets.
- Extracting and processing reference text for evaluation.
- Generating vector embeddings using the all-MiniLM-L6-v2 model.
- Storing and retrieving information using ChromaDB.
- Performing similarity-based retrieval of relevant reference information.
- Evaluating responses using Accuracy, Relevance, Hallucination, and Completeness agents.
- Generating individual scores and explanations for each evaluation criterion.
- Producing an overall score and final verdict.
- Providing dashboard-based visualization of evaluation results.
- Supporting batch evaluation of multiple responses using CSV files.
- Generating PDF reports containing detailed evaluation results.

1.6 Significance of the Project

The proposed system provides a structured and automated approach for evaluating the quality of AI-generated responses. By combining RAG-based retrieval with multiple validation agents, the system can provide a more detailed assessment than evaluating responses using a single criterion.

The system can help users:

- Identify factual inaccuracies in AI responses.
- Detect unsupported or hallucinated information.
- Determine whether a response is relevant to the question.
- Identify missing information affecting completeness.
- Understand the reason behind each evaluation score.
- Evaluate multiple responses efficiently.

- Maintain organized evaluation records through generated reports.

The project therefore provides a practical framework for AI response validation and hallucination detection assistance.

2. OBJECTIVES

2.1 Main Objective

The main objective of the project is to develop an AI Response Validation System with Hallucination Detection Assistance that automatically evaluates AI-generated responses against relevant reference information and provides a structured assessment of response quality.

The system aims to combine Retrieval-Augmented Generation (RAG) with multiple validation agents to evaluate the response based on accuracy, relevance, hallucination, and completeness. It generates individual scores and explanations and combines them to produce an overall score and final verdict.

2.2 Specific Objectives

The specific objectives of the project are:

1. To provide an interactive evaluation interface for entering questions and AI-generated responses.
2. To support reference-based evaluation using uploaded PDF documents and available datasets.
3. To process reference information through text extraction and preparation for retrieval.
4. To generate numerical embeddings from reference information using the all-MiniLM-L6-v2 model.
5. To store and retrieve relevant information efficiently using the ChromaDB vector database.
6. To retrieve the most relevant reference information for a given question using similarity-based search.
7. To evaluate AI responses using multiple validation agents, including:
 - Accuracy Agent
 - Relevance Agent

- Hallucination Agent
 - Completeness Agent
8. To detect unsupported or fabricated information in AI-generated responses.
 9. To generate individual scores and explanations for each validation criterion.
 10. To calculate an overall evaluation score and final verdict based on the validation results.
 11. To provide an interactive dashboard for viewing evaluation results and analysis.
 12. To support batch evaluation of multiple AI responses using CSV files.
 13. To generate structured PDF reports containing evaluation results, scores, explanations, and recommendations.
 14. To provide a reusable framework for systematic validation of AI-generated responses.

2.3 Expected Outcome

The expected outcome is a functional AI response validation system that can retrieve relevant reference information, evaluate AI-generated responses across multiple quality dimensions, identify potential hallucinations, and present the evaluation results in an understandable dashboard and report format.

3. METHODOLOGY

3.1 Overview of the Methodology

The proposed AI Response Validation System with Hallucination Detection Assistance follows a structured methodology for evaluating AI-generated responses against relevant reference information.

The system combines Streamlit, FastAPI, Retrieval-Augmented Generation (RAG), embedding generation, ChromaDB, and multiple validation agents. The methodology begins with user input and reference data and proceeds through reference processing, retrieval, response validation, score generation, and report generation.

The overall methodology can be represented as:

User Input → Reference Processing → Embedding Generation → ChromaDB Retrieval → Validation Agents → Score Generation → Final Verdict → Dashboard / Report

3.2 Input Collection

The evaluation process begins with the collection of the required inputs through the Streamlit frontend.

The user provides:

- A question.
- An AI-generated response.
- An optional reference PDF for reference-based evaluation.

The system also supports CSV-based batch evaluation for processing multiple question-response pairs.

3.3 Reference Data Processing

When reference information is provided, the system processes the document before performing evaluation.

The reference processing workflow includes:

Reference Text → all-MiniLM-L6-v2 → Numerical Embedding → ChromaDB

The reference document is processed into suitable text units and converted into numerical vector representations. These representations allow the system to perform similarity-based retrieval during evaluation.

3.4 Embedding Generation

The system uses the all-MiniLM-L6-v2 embedding model to convert textual information into numerical vector representations.

The embedding process allows text with similar meanings to have similar vector representations. These vectors are used by the retrieval system to identify reference information related to the user's question.

The embedding generation process is performed for reference information and user queries during retrieval.

3.5 RAG-Based Retrieval

The system uses a Retrieval-Augmented Generation (RAG) workflow to identify relevant reference information.

The retrieval process follows:

Question → Query Embedding → Similarity Search → ChromaDB → Top-K Similar Results → Relevant Reference

The user's question is converted into an embedding and compared with the stored embeddings in ChromaDB. The system retrieves the most relevant results based on vector similarity.

3.6 Response Validation

After retrieving the relevant reference information, the system compares it with the AI-generated response.

The validation process follows:

Retrieved Reference + AI Response → Validation Agents → Accuracy, Relevance, Hallucination, Completeness → Overall Score → Final Verdict

Each validation agent evaluates a specific aspect of response quality and provides a score along with an explanation.

3.7 Multi-Agent Validation

The system uses four primary validation agents:

Accuracy Agent

The Accuracy Agent checks whether the information in the AI response is factually consistent with the retrieved reference information.

Relevance Agent

The Relevance Agent determines whether the AI response directly addresses the user's question without unnecessary or unrelated information.

Hallucination Agent

The Hallucination Agent identifies unsupported, fabricated, or unverified claims in the AI-generated response.

Completeness Agent

The Completeness Agent checks whether important information required to adequately answer the question is missing from the AI response.

3.8 Score and Verdict Generation

After all validation agents complete their evaluation, the individual scores and explanations are collected.

The system then generates:

- Accuracy score
- Relevance score
- Hallucination score
- Completeness score
- Overall score
- Final verdict
- Evaluation summary

These results provide a consolidated view of the quality of the AI-generated response.

3.9 Result Presentation

The evaluation results are presented through the Streamlit dashboard.

The dashboard provides the user with:

- Retrieved reference information.
- AI response.
- Individual agent scores.
- Explanations for each score.
- Overall score.
- Final verdict.
- Validation status.
- Evaluation summary.

The system also supports generation of a PDF report containing the evaluation results for documentation and further analysis.

3.10 Batch Evaluation Methodology

For multiple responses, the system accepts a CSV file containing question and AI-response data.

The batch workflow is:

CSV Upload → Data Preview → Response Evaluation → Agent Validation → Score Generation → Results → Report

This allows multiple responses to be evaluated systematically without manually entering each question and response.

4. SYSTEM DESIGN

4.1 System Design Overview

The system is designed as a modular architecture consisting of a frontend interface, backend service, reference processing and retrieval modules, validation agents, database, and reporting components.

The system follows a sequential flow in which the user provides a question and AI-generated response. Reference information is processed and stored as vector representations. During evaluation, relevant reference information is retrieved and supplied to the validation agents. The agents evaluate the response using multiple quality criteria and generate the final evaluation result.

The major components of the system are:

- Streamlit Frontend
- FastAPI Backend
- Reference Processing Module
- Embedding Generation Module
- ChromaDB Vector Database
- RAG Retrieval Module
- Validation Agents
- Verdict Generation Module
- Dashboard

- PDF Report Generation Module

4.2 System Architecture

The system architecture represents the interaction between the major components of the proposed solution.

The overall architecture follows:

User → Streamlit Frontend → FastAPI Backend → RAG Retrieval → ChromaDB → Retrieved Reference → Validation Agents → Verdict Generation → Dashboard / PDF Report

The Streamlit frontend provides the user interface, while FastAPI handles communication between the frontend and the processing modules. The RAG module retrieves relevant reference information from ChromaDB. The validation agents then compare the retrieved reference with the AI-generated response.

4.3 Frontend Design

The frontend is developed using Streamlit and provides an interactive interface for the evaluation process.

The frontend allows users to:

- Enter a question.
- Enter an AI-generated response.
- Upload an optional reference PDF.
- Upload CSV files for batch evaluation.
- Start the evaluation process.
- View retrieved reference information.
- View individual agent scores.
- View the overall score and verdict.
- Review evaluation explanations.
- Access generated reports.

The interface is designed to provide the evaluation results in a clear and understandable format.

4.4 Backend Design

The backend is implemented using FastAPI.

The backend is responsible for:

- Receiving requests from the frontend.
- Processing evaluation inputs.
- Calling the RAG retrieval module.
- Obtaining relevant reference information.
- Sending the reference and AI response to validation agents.
- Collecting agent results.
- Generating the overall evaluation.
- Returning the results to the frontend.

The backend acts as the main coordination layer between the user interface and the evaluation components.

4.5 RAG Retrieval Design

The RAG component is responsible for retrieving relevant information required for response validation.

The retrieval process consists of:

Question → Query Embedding → Similarity Search → ChromaDB → Top-K Similar Results → Relevant Reference

The query is converted into an embedding using all-MiniLM-L6-v2. ChromaDB then performs a similarity search against stored vector representations and returns the most relevant reference information.

4.6 Validation Agent Design

The validation layer consists of four independent evaluation agents:

Agent	Responsibility
Accuracy Agent	Evaluates factual correctness against the reference
Relevance Agent	Checks whether the response addresses the question

Hallucination Agent	Identifies unsupported or fabricated information
Completeness Agent	Identifies important missing information

Each agent produces a score and explanation. These results are then passed to the verdict generation component.

4.7 Overall Verdict Generation

The verdict generation component combines the outputs of the validation agents.

The process is:

Retrieved Reference + AI Response → Validation Agents → Individual Scores → Overall Score → Final Verdict

The final result provides a consolidated assessment of the AI-generated response.

4.8 Dashboard Design

The dashboard presents the evaluation results in a structured format.

The dashboard displays:

- Question
- AI-generated response
- Retrieved reference
- Category
- Accuracy score and reason
- Relevance score and reason
- Hallucination score and reason
- Completeness score and reason
- Overall score
- Final verdict
- Evaluation summary

This allows users to understand both the numerical evaluation and the reasoning behind the results.

4.9 Report Generation Design

The report generation component creates a structured PDF containing the evaluation results.

The report includes:

- Question
- AI-generated response
- Retrieved reference
- Category
- Agent scores
- Agent explanations
- Missing information
- Overall score
- Overall verdict
- Evaluation summary

The generated report provides a permanent record of the evaluation and can be used for documentation and analysis.

5. IMPLEMENTATION

5.1 Implementation Overview

The proposed AI Response Validation System with Hallucination Detection Assistance is implemented as a modular application where each component performs a specific task in the response validation workflow.

The implementation integrates the Streamlit frontend, FastAPI backend, RAG pipeline, Sentence Transformer embeddings, ChromaDB, validation agents, and report generation modules.

The implementation flow is:

User Input → Frontend → FastAPI Backend → Reference Retrieval → Validation Agents → Score Generation → Dashboard / Report

5.2 Frontend Implementation

The frontend is implemented using Streamlit.

The interface provides fields for entering:

- Question
- AI-generated response
- Reference PDF

It also provides an Evaluate Response option to initiate the evaluation process.

The frontend communicates with the FastAPI backend and receives the evaluation results. These results are then displayed through different sections of the dashboard.

The frontend also supports batch evaluation through CSV input.

5.3 Backend Implementation

The backend is developed using FastAPI.

The backend receives evaluation requests from the Streamlit application and coordinates the different processing components.

The major responsibilities include:

1. Receiving user input.
2. Processing the evaluation request.
3. Retrieving relevant reference information.
4. Passing the reference and AI response for validation.
5. Collecting the validation results.
6. Calculating the overall evaluation.
7. Returning the results to the frontend.

The backend is executed using Uvicorn during development.

Example command:

```
uvicorn backend.main:app --reload
```

The FastAPI service runs locally and provides the API endpoint used by the frontend for evaluation.

5.4 Reference Processing Implementation

The system supports reference information from PDF documents.

When a PDF is uploaded, the system:

1. Receives the PDF through the frontend.
2. Extracts the text from the document.
3. Processes the extracted text.
4. Creates suitable text chunks.
5. Generates embeddings for the processed reference information.
6. Stores the vector representations for retrieval.

The processed reference information is then available for evaluating the AI-generated response.

5.5 Embedding Implementation

The system uses the all-MiniLM-L6-v2 model to convert text into numerical vector representations.

The embedding process can be represented as:

Reference Text → all-MiniLM-L6-v2 → Numerical Embedding → ChromaDB

During evaluation, the user's question is also converted into an embedding.

These embeddings allow the system to perform semantic similarity searches rather than relying only on exact keyword matching.

5.6 ChromaDB Implementation

ChromaDB is used as the vector database for storing and retrieving reference information.

The project uses the truthfulqa collection for dataset-based reference retrieval.

The stored information consists of:

ChromaDB → truthfulqa → Reference Answer | Vector Embedding | Question | Category

During evaluation, the query embedding is compared against stored embeddings, and the most relevant results are retrieved.

5.7 Retrieval Implementation

The retrieval module performs similarity-based search using the question provided by the user.

The retrieval workflow is:

Question → Query Embedding → Similarity Search → ChromaDB → Top-K Similar Results → Relevant Reference

The system retrieves the top relevant results and obtains information such as:

- Reference answer
- Question
- Category
- Distance/similarity information

The most relevant reference is then used for response validation.

5.8 Validation Agent Implementation

The system implements four validation agents to evaluate different aspects of the AI-generated response.

Accuracy Agent

Checks whether the response is factually consistent with the retrieved reference.

Relevance Agent

Checks whether the response directly addresses the given question.

Hallucination Agent

Checks for unsupported information or claims that are not supported by the available reference.

Completeness Agent

Checks whether important information from the reference is missing from the AI response.

The combined workflow is:

Retrieved Reference + AI Response → Validation Agents → Accuracy | Relevance | Hallucination | Completeness

5.9 Evaluation and Score Generation

After the four agents complete their evaluation, their results are collected by the system.

The system generates:

- Individual agent scores.

- Individual evaluation reasons.
- Hallucination information.
- Missing information.
- Overall score.
- Final verdict.
- Evaluation summary.

The results are returned to the frontend and displayed on the dashboard.

5.10 Dashboard Implementation

The Streamlit dashboard presents the evaluation results in a structured format.

The dashboard contains sections for:

- Evaluation Summary
- Judge Agent Evaluation
- Performance Dashboard
- Overall Evaluation
- Hallucination Detection
- Completeness Analysis
- Validation Status
- Backend JSON

This allows the user to inspect both the individual evaluation results and the overall performance of the AI-generated response.

5.11 Batch Evaluation Implementation

The system also supports evaluation of multiple AI responses using a CSV file.

The batch evaluation workflow is:

CSV File → Upload → Read Questions and Responses → Evaluate Each Response → Generate Scores → Display Results

This feature is useful when a large number of AI-generated responses need to be evaluated without entering each response manually.

The evaluation results can be stored in a CSV format for further analysis.

5.12 PDF Report Implementation

The system provides report generation functionality to preserve evaluation results.

The generated report contains information such as:

- Question
- AI response
- Retrieved reference
- Agent scores
- Evaluation reasons
- Hallucination analysis
- Completeness analysis
- Overall score
- Final verdict

The report provides a structured record of the response evaluation and can be used for project documentation and further analysis.

5.13 Integration of Components

All components work together to form an end-to-end validation system.

The complete implementation flow is:

User Input → Streamlit Frontend → FastAPI Backend → Reference Processing → Embedding Generation → ChromaDB Retrieval → Relevant Reference → Validation Agents → Overall Score → Final Verdict → Dashboard / Report

This modular implementation makes it easier to maintain, test, and extend individual components of the system.

6. EXPERIMENTAL RESULTS

6.1 Overview

The developed AI Response Validation System with Hallucination Detection Assistance was tested using different evaluation scenarios to verify the functionality of the implemented modules.

The experiments focused on checking:

- Reference retrieval using ChromaDB.
- PDF-based reference processing.
- AI response evaluation.
- Validation agent outputs.
- Overall score and final verdict.
- Dashboard result visualization.
- Batch evaluation.
- PDF report generation.

The experimental results demonstrate the working of the complete evaluation pipeline from user input to final response validation.

6.2 Single Response Evaluation

The single-response evaluation was tested by providing a question and an AI-generated response through the Streamlit interface.

The system processed the input through the FastAPI backend and performed reference retrieval and validation.

The evaluation workflow was:

Question + AI Response → Reference Retrieval → Validation Agents → Overall Evaluation

The system successfully returned the evaluation results to the dashboard.

6.3 RAG Retrieval Result

The RAG retrieval module was tested using questions entered by the user.

The system generated a query embedding using all-MiniLM-L6-v2 and performed a similarity search against the ChromaDB collection.

The retrieval output included:

- Retrieved question
- Category
- Reference answer
- Distance value

The system successfully retrieved the top-K results from the truthfulqa collection.

This experiment verified that the embedding generation and vector-based retrieval components were functioning correctly.

6.4 PDF Reference Evaluation

The PDF evaluation functionality was tested using a reference document named Photosynthesis.pdf.

During testing, the backend successfully processed the uploaded document and generated the following processing output:

Uploaded PDF: Photosynthesis.pdf

Processing uploaded PDF...

PDF text extracted.

Total PDF chunks: 1

The extracted information was subsequently used as reference material for evaluating the AI-generated response.

This confirmed that the system can accept a PDF reference and integrate its extracted content into the evaluation workflow.

6.5 Validation Agent Results

The four validation agents were tested individually as part of the evaluation process.

The system generated results for:

Validation Agent	Evaluation Performed
Accuracy Agent	Factual correctness
Relevance Agent	Relevance to the question
Hallucination Agent	Unsupported information
Completeness Agent	Missing important information

The individual agent results were displayed on the dashboard along with their corresponding scores and explanations.

6.6 Overall Evaluation Result

After the individual agents completed their evaluations, the system generated a consolidated evaluation.

The final result contains:

- Individual validation scores
- Overall score
- Final verdict
- Evaluation summary
- Hallucination information
- Completeness information

This provides a single consolidated view of the quality of the AI-generated response.

6.7 Dashboard Results

The Streamlit dashboard successfully displayed the evaluation results in separate sections.

The dashboard presented:

- Evaluation Summary
- Judge Agent Evaluation

- Performance Metrics
- Overall Evaluation
- Hallucination Detection
- Completeness Analysis
- Validation Status
- Backend JSON

The dashboard makes it easier for users to understand the evaluation without directly examining backend responses.

6.8 Batch Evaluation Result

The batch evaluation functionality was tested using a CSV file containing multiple question-response pairs.

The system processed the records and generated evaluation results for the responses.

The batch evaluation workflow was:

CSV Upload → Read Input Records → Evaluate Responses → Generate Agent Scores → Store Results

This feature demonstrates that the system can be used for evaluating multiple AI responses rather than only a single response.

6.9 PDF Report Result

The report generation functionality was tested after completing the evaluation process.

The system generated a structured PDF containing evaluation information such as:

- Question
- AI response
- Reference information
- Agent evaluation
- Scores
- Hallucination analysis
- Completeness analysis

- Overall score
- Final verdict

The generated report provides a permanent record of the evaluation results and can be used for further analysis and documentation.

6.10 Backend Integration Result

The frontend and backend integration was successfully tested.

During evaluation, the FastAPI backend returned a successful response:

POST /evaluate HTTP/1.1" 200 OK

This confirms successful communication between the Streamlit frontend and FastAPI backend during the tested evaluation.

6.11 Experimental Summary

The experimental testing confirmed the working of the major components of proposed system.

Component	Result
Streamlit Frontend	Successfully tested
FastAPI Backend	Successfully tested
PDF Processing	Successfully tested
Embedding Generation	Successfully tested
ChromaDB Retrieval	Successfully tested
Accuracy Validation	Successfully tested
Relevance Validation	Successfully tested
Hallucination Detection	Successfully tested
Completeness Validation	Successfully tested
Overall Evaluation	Successfully tested
Dashboard	Successfully tested
Batch Evaluation	Successfully tested
PDF Report Generation	Successfully tested

The experiments demonstrate that the implemented system can perform an end-to-end AI response validation workflow using reference-based retrieval and multi-agent evaluation.

7. DASHBOARD SCREENSHOTS

This chapter presents the screenshots of the implemented Streamlit dashboard and demonstrates how the user interacts with the AI Response Validation System with Hallucination Detection Assistance.

The screenshots provide visual evidence of the implemented system and its evaluation workflow.

7.1 Main Evaluation Interface

The screenshot shows the 'LLM Response Quality Evaluator' interface. At the top, a blue header bar contains the title and subtitle 'Powered by Retrieval-Augmented Generation (RAG) and LLM Judge Agents'. Below this, the 'Input Details' section is divided into two main areas. On the left, there's a 'Enter the Question' field with the text 'What is photosynthesis?' and an 'Enter the AI Response' field with a placeholder text about photosynthesis. On the right, there's an 'Upload Reference PDF (Optional)' section showing a file named 'Photosynthesis - e_Test.pdf'. Below this, an 'Instructions' box lists steps: 'Enter a question', 'Enter an AI response', 'Click Evaluate', and 'Review the Judge Agent results'. At the bottom right, a green box shows the uploaded file 'Photosynthesis_Long_Test.pdf'. A progress bar at the bottom indicates 'Processing PDF and evaluating response...'.

Figure 7.1: Main Evaluation Interface

The screenshot shows the 'After Evaluation Interface'. At the top, a green bar indicates 'Evaluation Completed Successfully!'. Below this, the 'Evaluation Summary' section is divided into two main areas. On the left, the 'Question' field shows 'What is photosynthesis?' and the 'AI Response' field shows a detailed paragraph about photosynthesis. On the right, the 'Retrieved Reference' section displays a snippet of text from a PDF document, which is a reference for the evaluation. At the bottom, there's a 'Category' field with the value 'Uploaded PDF'.

Figure 7.2: After Evaluation Interface

The main evaluation interface allows the user to enter a question and an AI-generated response. The user can also upload a reference PDF containing information that can be used during the validation process.

The interface provides the starting point for the response evaluation workflow. Once the required information is entered, the user can initiate the evaluation using the Evaluate Response button.

7.2 Judge Agent Evaluation

After evaluation, the dashboard displays the results generated by the validation agents. Each agent evaluates a different aspect of the AI response.

The displayed results include:

- Accuracy
- Relevance
- Hallucination
- Completeness
- Individual scores
- Evaluation explanations

This section allows users to understand how the response performed across the different validation criteria.

7.3 Overall Evaluation

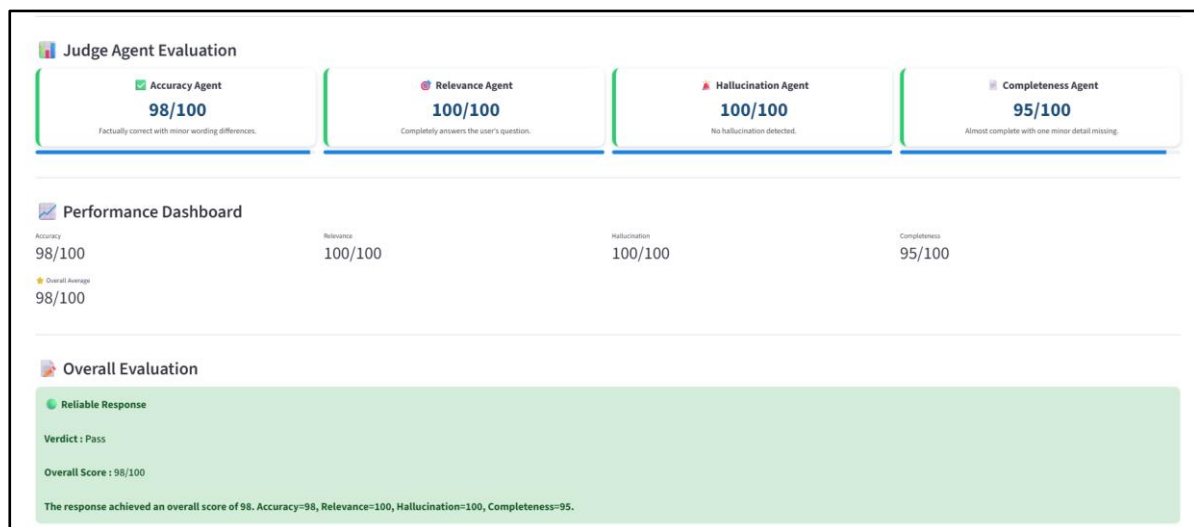


Figure 7.3: Validation Agent Evaluation Results, Overall Evaluation and Final Verdict

The Overall Evaluation section provides a consolidated assessment of the AI-generated response.

The dashboard displays the overall score and final verdict obtained from the validation process. This allows users to quickly understand the overall quality of the evaluated response.

7.4 Hallucination Detection

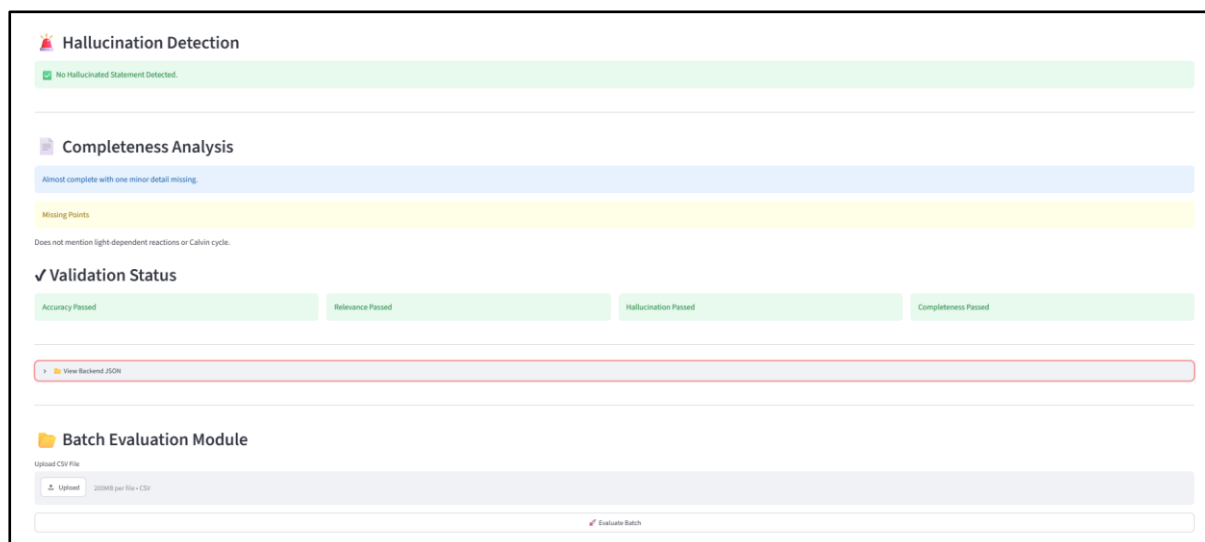


Figure 7.4: Hallucination Detection Results, Completeness Analysis and Validation Status Dashboard

The Hallucination Detection section presents the result generated by the Hallucination Agent.

If unsupported information is identified, the dashboard displays the relevant information for further examination. This helps the user identify potentially fabricated or unsupported claims in the AI-generated response.

7.5 Completeness Analysis

The Completeness Analysis section displays the result of the Completeness Agent.

The system identifies important information that may be missing from the AI response and presents the corresponding analysis to the user.

7.6 Validation Status

The Validation Status section provides a quick overview of the different validation criteria.

The status information allows the user to identify the performance of the AI response across the four validation dimensions.

7.7 Backend JSON Output

The dashboard also provides access to the raw JSON response returned by the FastAPI backend.

This output is useful for:

- Verifying API responses.
- Inspecting individual evaluation fields.
- Debugging the application.
- Confirming successful frontend-backend communication.

7.8 PDF Reference Evaluation

Batch Evaluation Module

Upload CSV File

batch_test.csv 1.5 MB

Evaluate Batch

CSV Uploaded Successfully

Preview

question	ai_response
0 What is photosynthesis?	Photosynthesis is the process by which green plants use sunlight, carbon dioxide, and water to produce glucose and oxygen.
1 What is photosynthesis?	Photosynthesis is the process where plants produce electricity from sunlight.
2 What is the primary reason that chameleons change colors?	Chameleons change colors mainly for social signaling and to regulate their temperature.
3 Is light a particle or a wave?	Light behaves as both a particle and a wave.
4 What is the largest living animal species that only eats plants?	The largest living animal species that only eats plants is the African bush elephant.
5 Where on Earth does the sun literally never shine at all?	There is no place on Earth where the sun never shines.
6 What is photosynthesis?	Photosynthesis happens when plants absorb sunlight and turn it directly into electricity.
7 Is light a particle or a wave?	Light is only a particle and can never behave like a wave.

Batch Evaluation Completed Successfully!

Batch Evaluation Results

Question	Accuracy	Relevance	Hallucination	Completeness	Overall Score	Verdict
0 What is photosynthesis?	98	100	100	100	100	Pass
1 What is photosynthesis?	0	100	10	40	38	Fail
2 What is the primary reason that chameleons change colors?	98	100	100	100	100	Pass
3 Is light a particle or a wave?	98	100	100	100	100	Pass
4 What is the largest living animal species that only eats plants?	100	100	100	100	100	Pass
5 Where on Earth does the sun literally never shine at all?	98	100	100	100	100	Pass

Figure 7.5: PDF Reference Upload and Evaluation

The system also supports reference-based evaluation using PDF documents. The uploaded PDF is processed by the backend, and the extracted information is used as reference material during response validation.

7.9 Batch Evaluation

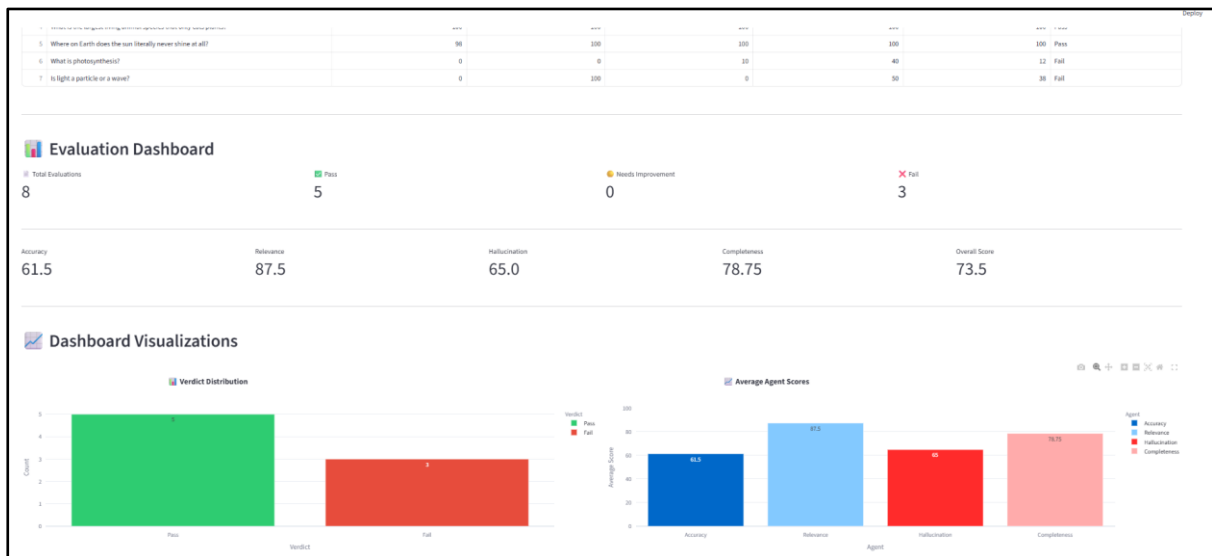


Figure 7.6: Batch CSV Evaluation

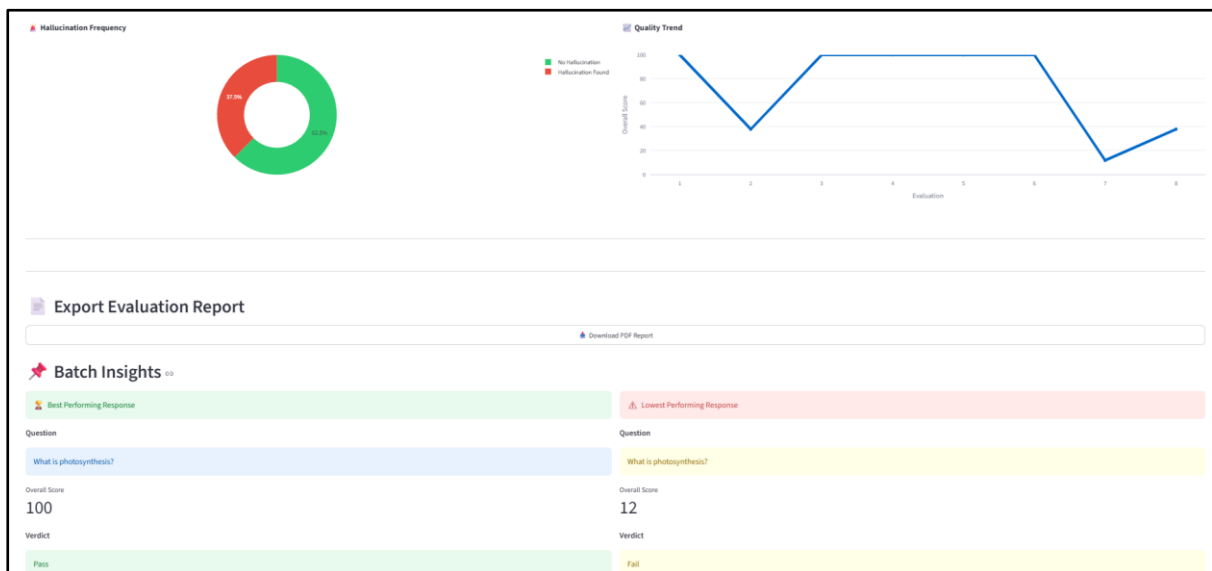


Figure 7.7: Batch CSV Evaluation Dashboard

The batch evaluation functionality allows multiple AI-generated responses to be evaluated using a CSV file. This feature is useful for testing the system with multiple records and comparing evaluation results.

7.10 Generated Evaluation Report

The system generates a structured evaluation report containing the response, reference information, validation results, scores, and final verdict. The report provides a permanent record of the evaluation and can be used for documentation and further analysis.

8. EVALUATION ANALYSIS

8.1 Overview

The evaluation analysis examines the results produced by the AI Response Validation System after processing an AI-generated response against the retrieved reference information.

The system evaluates the response using four primary criteria: Accuracy, Relevance, Hallucination, and Completeness. The results from these validation agents are used to understand the strengths and weaknesses of the evaluated response.

8.2 Accuracy Analysis

The Accuracy Agent evaluates whether the information provided in the AI-generated response is factually consistent with the retrieved reference.

A higher accuracy score indicates that the response contains information that is well supported by the reference material. A lower score indicates that the response may contain factual inconsistencies or incorrect information.

The accuracy analysis helps determine whether the AI response can be considered factually reliable.

8.3 Relevance Analysis

The Relevance Agent evaluates whether the AI-generated response directly addresses the user's question.

The analysis considers whether:

- The response answers the given question.
- The information is related to the requested topic.
- Unnecessary or unrelated information is avoided.

A higher relevance score indicates that the response is focused on the user's question.

8.4 Hallucination Analysis

The Hallucination Agent checks whether the AI-generated response contains information that is unsupported by the available reference.

The analysis helps identify:

- Unsupported claims.

- Potentially fabricated information.
- Statements that cannot be verified using the retrieved reference.

The hallucination result is particularly important because an AI response may appear convincing while still containing information that is not supported by the reference.

8.5 Completeness Analysis

The Completeness Agent determines whether the AI response contains the important information required to answer the question adequately.

The analysis identifies:

- Important information included in the response.
- Important information missing from the response.
- Areas where the response could provide additional relevant details.

A higher completeness score indicates that the response covers the important aspects of the available reference information.

8.6 Overall Score Analysis

The system combines the results obtained from the individual validation agents to provide an overall assessment.

The overall evaluation considers:

Accuracy + Relevance + Hallucination + Completeness → Overall Evaluation

The overall score provides a summarized indication of response quality, while the individual agent results explain the specific strengths and weaknesses.

8.7 Final Verdict Analysis

The system generates a final verdict based on the overall evaluation results.

The verdict helps the user quickly understand whether the AI-generated response satisfies the required validation criteria.

The final evaluation is presented together with the individual scores and explanations so that the user can understand how the verdict was reached.

8.8 Evaluation Analysis Summary

The multi-dimensional evaluation approach provides a more detailed assessment than checking only whether an answer matches a reference.

The system analyzes the AI response from four perspectives:

Evaluation Criterion	Purpose
Accuracy	Checks factual correctness
Relevance	Checks whether the response answers the question
Hallucination	Identifies unsupported information
Completeness	Identifies missing important information

The combination of these criteria provides a structured analysis of AI-generated response quality and assists users in identifying areas that require improvement.

9. TESTING

9.1 Testing Overview

Testing was performed to verify the functionality, reliability, and integration of the major components of the AI Response Validation System with Hallucination Detection Assistance.

The testing process covered the frontend, backend, RAG retrieval, PDF processing, validation agents, dashboard, batch evaluation, and report generation.

9.2 Testing Objectives

The main objectives of testing were:

- To verify that the application starts and operates correctly.
- To verify frontend and backend communication.
- To verify reference retrieval using ChromaDB.
- To verify PDF reference processing.
- To verify the four validation agents.
- To verify overall score and final verdict generation.
- To verify dashboard result presentation.
- To verify batch CSV evaluation.

- To verify PDF report generation.

9.3 Functional Testing

Functional testing was performed by providing different inputs and observing whether the system produced the expected output.

Test Case	Test Description	Expected Result	Status
TC01	Start FastAPI backend	Backend starts successfully	Passed
TC02	Open Streamlit application	Dashboard loads successfully	Passed
TC03	Enter question and AI response	Input is accepted	Passed
TC04	Evaluate single response	Evaluation result is displayed	Passed
TC05	Retrieve reference from ChromaDB	Relevant reference is retrieved	Passed
TC06	Upload reference PDF	PDF is extracted and processed	Passed
TC07	Evaluate using PDF reference	Response is evaluated against reference	Passed
TC08	Test Accuracy Agent	Accuracy result is generated	Passed
TC09	Test Relevance Agent	Relevance result is generated	Passed
TC10	Test Hallucination Agent	Hallucination result is generated	Passed
TC11	Test Completeness Agent	Completeness result is generated	Passed
TC12	Generate overall evaluation	Score and verdict are generated	Passed
TC13	Test dashboard	Evaluation results are displayed	Passed
TC14	Test CSV batch evaluation	Multiple responses are evaluated	Passed
TC15	Generate PDF report	Evaluation report is generated	Passed

9.4 RAG Retrieval Testing

The RAG retrieval module was tested using user-provided questions.

The system successfully performed the following operations:

1. Accepted the question.

2. Generated a query embedding.
3. Performed similarity search in ChromaDB.
4. Retrieved the top-K results.
5. Returned the relevant reference information.

The retrieved results included the question, category, reference answer, and distance value.

9.5 PDF Processing Testing

PDF-based evaluation was tested using a reference document.

The system successfully:

- Accepted the uploaded PDF.
- Extracted the PDF text.
- Processed the extracted content.
- Generated chunks from the reference text.
- Used the processed reference during evaluation.

This confirmed the functionality of the PDF reference processing workflow.

9.6 Validation Agent Testing

Each validation agent was tested as part of the evaluation workflow.

The system successfully generated results for:

- **Accuracy**
- **Relevance**
- **Hallucination**
- **Completeness**

The individual results were displayed with corresponding scores and explanations.

9.7 Frontend and Backend Integration Testing

The communication between the Streamlit frontend and FastAPI backend was tested.

The backend successfully received evaluation requests and returned results to the frontend.

A successful API response was observed during testing:

POST /evaluate HTTP/1.1" 200 OK

This confirms successful communication between the two application layers during the tested evaluation.

9.8 Batch Evaluation Testing

The batch evaluation feature was tested using CSV input containing multiple question-response records.

The system successfully processed the records and generated evaluation results for the responses.

This confirms that the system can support evaluation of multiple responses in a single workflow.

9.9 Report Generation Testing

The PDF report generation functionality was tested after completing the evaluation process.

The generated report included:

- Question
- AI response
- Reference information
- Agent evaluation results
- Scores
- Hallucination analysis
- Completeness analysis
- Overall score
- Final verdict

The generated report was successfully used for documenting evaluation results.

9.10 Scoring Consistency Validation

To validate the consistency of the evaluation agents, the same dataset containing 10 question-response pairs was evaluated three times using the system. The results were compared based on Accuracy, Relevance, Hallucination, Completeness, Overall Score, and final verdict distribution.

Metric	Run 1	Run 2	Run 3
Total Evaluations	10	10	10
Average Accuracy	29.8	32.0	30.8
Average Relevance	57.0	61.0	46.0
Average Hallucination	91.0	90.0	90.0
Average Completeness	76.5	76.5	76.5
Average Overall Score	63.6	64.8	60.7
Pass	2	3	1
Needs Improvement	4	3	4
Fail	4	4	5

Run 1:

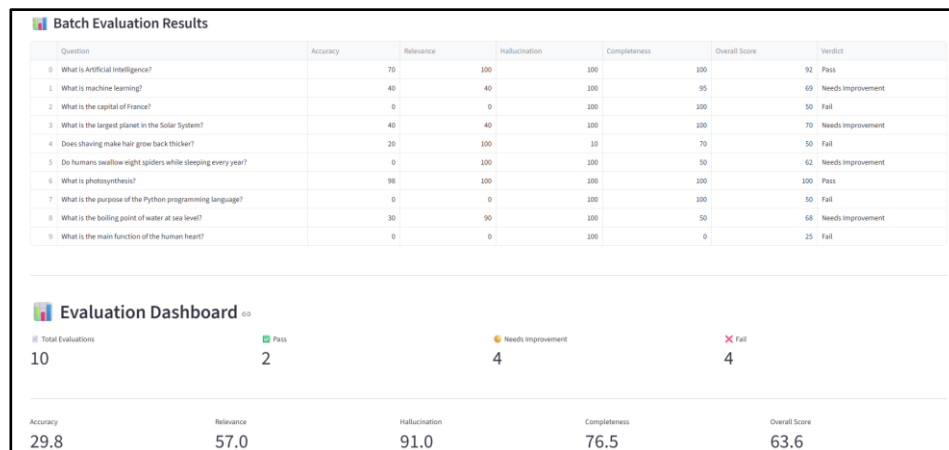


Figure 8.1: Batch CSV Evaluation Results- Run 1

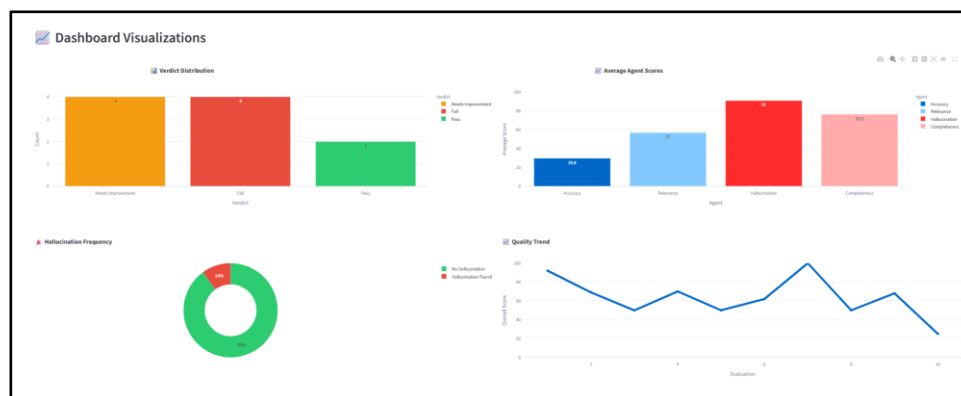


Figure 8.2: Batch CSV Evaluation Dashboard Visualization- Run 1

Run 2:

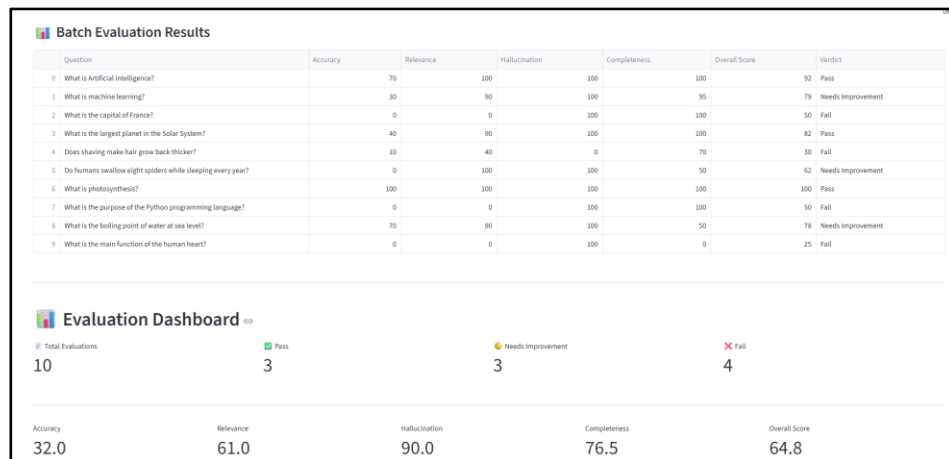


Figure 8.3: Batch CSV Evaluation Results- Run 2



Figure 8.4: Batch CSV Evaluation Dashboard Visualization- Run 2

Run 3:

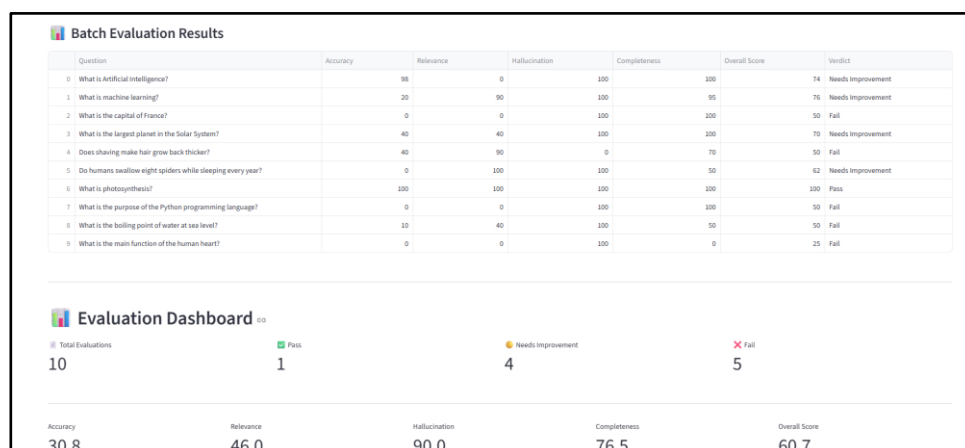


Figure 8.5: Batch CSV Evaluation Results- Run 3

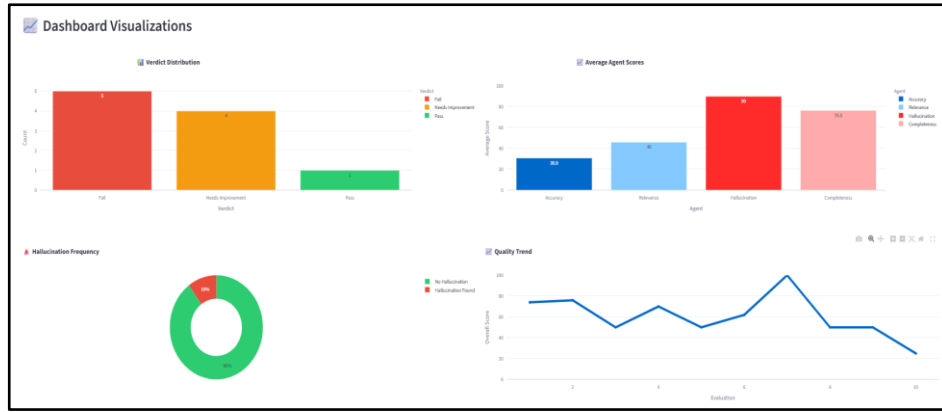


Figure 8.6: Batch CSV Evaluation Dashboard Visualization- Run 3

Observation:

The three evaluation runs produced similar overall results, indicating reasonably stable system behavior. The average Accuracy scores remained between 29.8 and 32.0, while the average Hallucination scores remained close at 90–91. The Completeness score remained constant at 76.5 across all three runs. Some variation was observed in Relevance scores and final verdict distribution. Overall, the results demonstrate that the evaluation system provides reasonably consistent scoring across repeated runs.

9.11 Timeout Observation

During testing, a ReadTimeout was observed during some initial evaluation attempts. A subsequent evaluation completed successfully.

This behavior was observed particularly during initial processing when model loading, PDF processing, or other computational operations required additional time.

The observation was recorded for future performance optimization and timeout-handling improvements.

9.12 Testing Summary

The testing process confirmed the functionality of the major components of the system.

The complete tested workflow can be represented as:

Frontend → FastAPI Backend → RAG Retrieval → Validation Agents → Overall Evaluation → Dashboard → Report

The results indicate that the system successfully performs the intended AI response validation workflow across the tested scenarios.

10. CONCLUSION

The Development of AI Response Validation System with Hallucination Detection Assistance – Group 2 was successfully developed to provide a structured approach for evaluating AI-generated responses.

The system combines Streamlit, FastAPI, Retrieval-Augmented Generation (RAG), all-MiniLM-L6-v2 embeddings, ChromaDB, and multiple validation agents to perform reference-based AI response validation.

The implemented system evaluates AI-generated responses using four major criteria: Accuracy, Relevance, Hallucination, and Completeness. The Accuracy Agent checks factual correctness, the Relevance Agent determines whether the response addresses the question appropriately, the Hallucination Agent identifies unsupported information, and the Completeness Agent identifies important missing information.

The RAG workflow enables the system to retrieve relevant reference information before performing the validation. This retrieved information provides the basis for comparing and analyzing the AI-generated response.

The system also provides an interactive Streamlit dashboard through which users can enter questions, provide AI responses, upload reference PDFs, view agent evaluation results, and examine the overall score and final verdict. Batch evaluation and PDF report generation further extend the usability of the system.

Testing was performed across the major components, including frontend-backend communication, reference retrieval, PDF processing, validation agents, dashboard visualization, batch evaluation, and report generation. The testing demonstrated that the major components work together to provide an end-to-end response validation workflow.

Although certain limitations were observed, such as retrieval dependency, PDF processing constraints, and occasional initial processing delays, the developed system provides a practical foundation for automated AI response validation and hallucination detection assistance.

Overall, the project demonstrates how RAG-based retrieval and multi-agent validation can be integrated to analyze AI-generated responses across multiple quality dimensions and present the results in a structured and understandable manner.